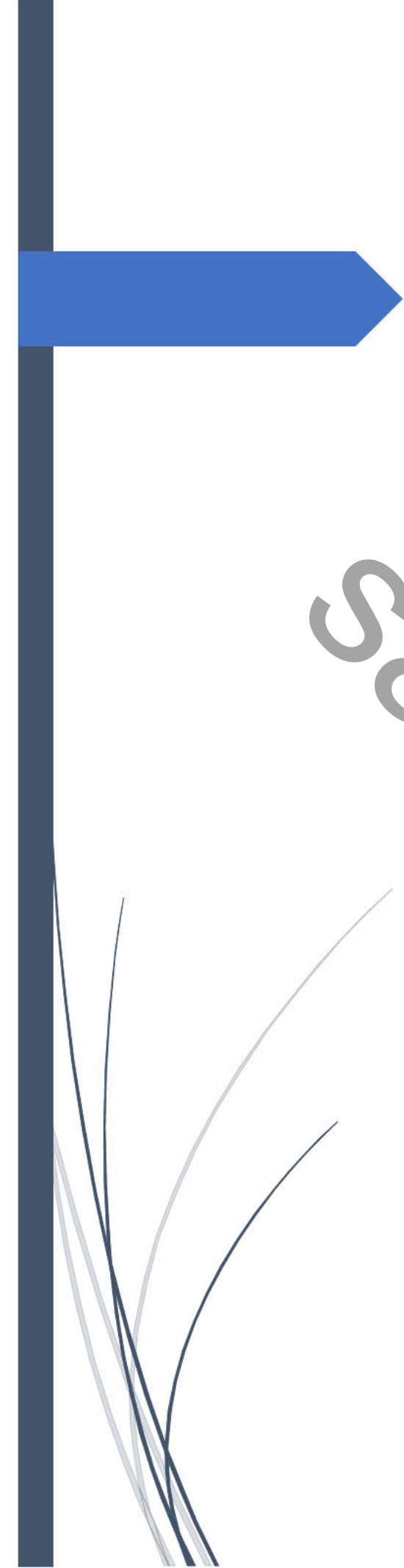




ANALYTICS FOR MANAGER

Data analysis and data-driven
decisions.

sample

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Introduction

In today's era, digital technology has changed the way of conducting organisational work based on the business-driven strategy as an enormous number of data is being collected by different industries. Technological devices such as laptops and smartphones allow organisations and their stakeholder to create and store a large amount of data of their customers. Furthermore, these large industries' data is analysed in different sectors for business decision-making to gain a competitive advantage in various business areas (Goar & Yadav, 2022). Data analytics is used by multiple industries for decision-making, such as healthcare, banking, education, communication, retail, and transportation (Le & Liaw, 2017). In this fourth industrial revolution, the Internet of Things – IoT (physical device network transfer data from one device to another without human intervention) enable retailers to improve their decision in customer experience, supply chain management, new customer segment revenue generation, pricing, and discounting strategy, and in-store customer behaviour analysis. Moreover, with the help of data analysis, a retailer, wholesaler, and manager can analyse the market trend and set their marketing strategy to create a large customer base (Matthew et al., 2015).

This report aims to analyse the dataset and make a data-driven decision in the retail superstore supplying office equipment product. Moreover, the dataset is selected from Kaggle, and various scholarly journals and articles are selected from the Robertson Library to support the data analysis results and the arguments. This report will also focus on three major business problems based on the selected dataset: product analysis, pricing and discounting, and logistic/transport preference. Retail superstore has an extensive range of product under various brands; thus, it is worth getting insight into data analytics tools and how it is helpful in the retail sector for data-driven decision-making. Furthermore, in this report, dataset analysis helps understand how data analytics allows retailers of office supplies accessories to make decisions in crucial business areas such as discount, pricing, logistic preference, and product-centric and customer-centric business model. Further, this report will discuss the data description of the selected dataset, data analysis method, and visualisation with the recommended data-driven decision for the manager.

Data Description

Different industries collect a large volume of data daily, which is difficult to analyse as raw data. Data collected from various sectors are characterised based on volume (size of data), variety(different types of data), and velocity(a dynamic form of data) of data which is difficult

to manage because of the complexity of data (Black, 2023). All numerical data are not analysed similarly because different numbers represent different entities. Major types of data measurement for the bifurcation of numerical data, which are discussed below (Black, 2023).

1. Nominal: This is the lowest level of data measurement, and it can only be used for classifying or categorising.
2. Ordinal: This data measurement gives the rank or order for people or objects.
3. Interval: This data measurement measures the distance between numbers when all the data are in numbers.
4. Ratio: The highest data measurement level represents the distance among the values.

This report selects the superstore dataset for the data-driven decision-making process and solving business issues in contemporary e-commerce and retailers of office and technology products supply organisations. This data is chosen from Kaggle, which is related to the superstore customer data in three major product categories: Furniture, Technology products, and Office supplies (). The selected data set is processed and cleaned before using it in the RapidMiner for predictive analysis, described in Figure 1 and categories are assigned with numbers. This report uses distinctive Microsoft software, such as Power BI for visualisation, Excel for descriptive and diagnostic analysis, and RapidMiner for predictive analysis. Different attributes of the datasets are summarised below with descriptions and categorisations.

1. Product Analysis for low-performing product line

Retail stores have multiple product range under different brands. In the retail sector, it is crucial to focus the product to analyse the best-performing and least-performing products, product-making losses, and consistency of the individual product performance (Saxena, 2021). Product analysis is crucial for Managers to make data-based decisions about which products to be continued, which products need to stock more, and which are the least and best-performing products. Below is a graphical representation of products with good profits and products that incur losses in the selected dataset. Further, diagnostic analysis is done for the brand and type of the product which incurs the loss.

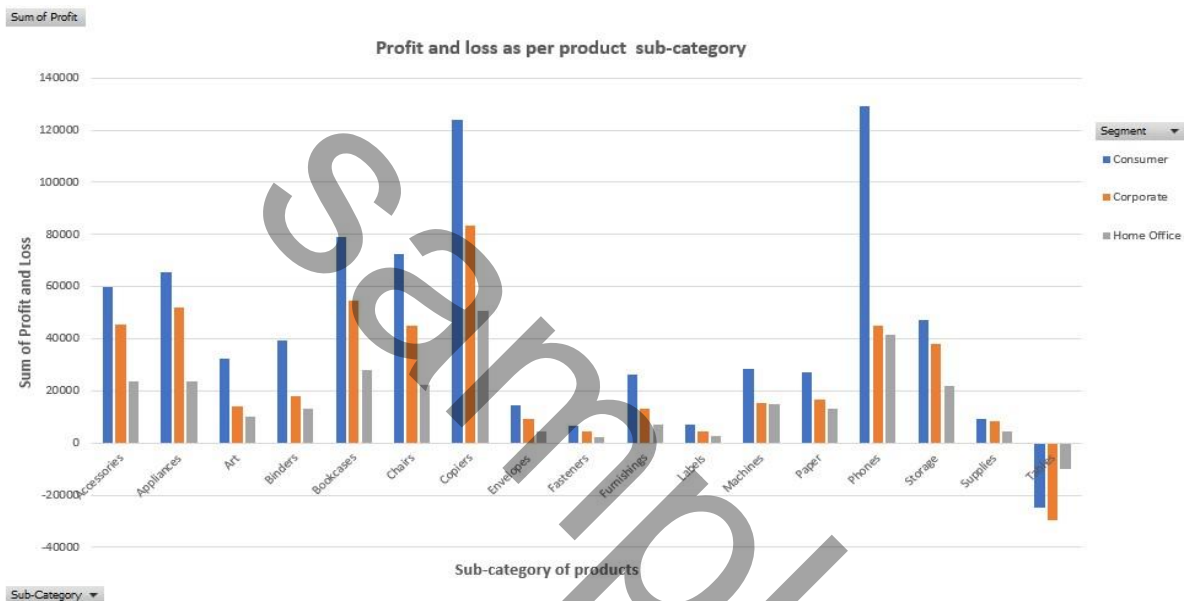


Figure 2: Product analysis as per customer segments

The above bar graph represents revenue according to different product performances in different segments. As per Figure 2, the highest profit-making products are phones and copier machines in the consumer segment. However, per the product analysis, tables are under loss in consumer, corporate, and home office segments, with the highest loss in the corporate segment. Managers can make a data-based decision on the product analysis by analysing the reason for failure in the product. Furthermore, the below graphs represent a detailed analysis of the loss-making product with an analysis of variables that causes the loss for specific products and loss-making brands.

Figure 3 demonstrate discount available on the product for sub-categories of the product line of the superstore. The highest discount is available on the tables and the minor discount on papers and art-related products. Thus, the discount might be the reason for the loss in the tables. Furthermore, Figure 4 demonstrates that the highest discount is available for conference, assembled, adjustable, and rectangular tables. Besides this, underperforming brands are Chromcraft, Bevis, Lerso, and Barricks, demonstrated as per world cloud in Figure 4. As a result, the most loss-making tables type are conference, adjustable, assembled, and rectangular tables under various brands such as Chromcraft, Bavis, and Barrick, respectively. To add to this, managers or retailers need to investigate why such a high discount is given for a specific product compared to the other product line. As per Figure 5, diagnostic analysis demonstrates that the number of quantities sold for the tables is deficient in all customer segments; thus, discounts for the specific brand and type of tables may be increased to boost the table retailer sales. Moreover, the chi-square test result indicates that the product category and customer segments influence each other. However, the consumer segment performs well compared to others by popular sales of specific product lines. Additionally, another diagnostic analysis suggests a significant difference in the discount available for a table in three different segments as per the figure below of the ANOVA test.

Anova: Single Factor						
SUMMARY						
<i>Groups</i>	<i>Count</i>	<i>Sum</i>	<i>Average</i>	<i>Variance</i>		
Consumer	158	4E+05	2414.6962	4388779.19		
Corporate	132	2E+05	1861.03788	2759200.34		
Home Office	91	1E+05	1422.42857	1260039.87		
ANOVA						
<i>Source of Variation</i>	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>	<i>P-value</i>	<i>F crit</i>
Between Groups	60001313.9	2	30000657	9.74334216	7.4814E-05	3.01960004
Within Groups	1163897167	378	3079093.03			
Total	1223898480	380				

Figure 7: Diagnostic analysis (ANOVA) for a significant difference in a discount for tables

Discount is not the only cause for the loss in the table product for specific brands in the superstore. Additionally, another variable also influences the brand's performance, such as the type and design of the product, customer satisfaction level, market competition, demand for

the products, etc. Similarly, according to Lal & Matutes (1994), retailers use marketing tools such as advertising, price reduction, and reducing delivery time and shipping cost to attract more customer base and increase sales. Additionally, customers are more likely to be concerned about the delivery time in the retail sector. It also adversely affects product demand and incurs additional costs to the company, which is reflected as the cause of the loss. Arguably, Kumar et al., (2017) emphasised that retailers should use different strategies at the market(competitive strategy), firm, store, and customer level to increase profitability. He has done an empirical analysis of variables such as products of various brands, unit cost, transportation cost, customer satisfaction(rating), and competitors' performance to analyse the overall profitability of the retail firm. Market-level and firm-level strategies such as market expansion and multichannel marketing are crucial for the retailer, which help retailers provide the latest products per the market trend and customer requirements (Kumar et al., 2017).

To make a data-driven decision, a manager or retailer must investigate why the highest discount is given in the specific brand and product line. Besides this, the manager should focus on the marketing strategy, customer satisfaction level, market trend and product demand, and inventory management of low-performing brands to mitigate the loss in the furniture category. Moreover, market trends and customer buying pattern analysis help the retailer to analyse the current market competitiveness.

2. Pricing and discounting strategy

Setting appropriate pricing for the target consumer is one of the critical decisions for retail profitability and customer segmentation. Retailers frequently use pricing discrimination to boost sales and use price discounts or quantity discounts (Khan & Jain, 2005). In today's digitalised world, e-commerce retailers also use advanced technology to deliver the right pricing offer to the right customer at the right time. Similarly, pricing analysis helps managers analyse specific patterns or trends in customer buying behaviour, which helps retailers increase profitability by providing appropriate prices (Bodea & Ferguson, 2014). Moreover, appropriate discount helps the manager to analyse buying behaviour and customer engagement. In the retail management context, business issues can arise that what could be the best promotion strategy to deliver high customer satisfaction and good revenue because discounting and price reduction are negatively correlated to the firm's income which is presented below in Figure 8.

	Sales/order value	Quantity in the order	Discount	Profit and loss	Shipping Cost	Delivery Days
Sales/order value	1					
Quantity in the order	0.31358796	1				
Discount	-0.086713171	-0.019874695	1			
Profit and loss	0.485158688	0.104427656	-0.314744871	1		
Shipping Cost	0.767827967	0.271757119	-0.078064594	0.354587293	1	
Delivery Days	-0.004657511	0.004346988	-0.001507438	0.001754576	-0.142541663	1

Figure 8: Predictive Analysis (Co-relation)

As the above correlation shows, sales amount and discounts are negatively correlated as discount increases and sales order value decreases (-0.08). Additionally, a high discount negatively impacts profit because a higher discount lowers the profit. Moreover, a reduced pricing strategy also increases the order quantity from regular sales to the amount of product. Thus, appropriate discounts and pricing will only sometimes cause loss or reduce profit. Similarly, as demonstrated in predictive analysis, shipping cost is highly correlated with sales order value (0.767), which means increasing order value will lead to higher shipping costs. However, a high correlation between shipping cost and sales order value can only be predicted for some products and sales orders as many other variables affect shipping cost per order, such as shipping priority, order priority, delivery days, and types of products. To add to this, it is noted that there is no significant correlation between delivery days and profit in retail / E-commerce superstore, which demonstrate that customer probably would not care about delivery time. However, in this fourth industrial revolution, it is the prime responsibility of retailers to ensure that product delivery is timely because it may cause additional costs to the company. Furthermore, the correlation of quantity in the order and sales value is positively related as quantity increases, sales value increases, but this is not applicable in all the office supplies retailing as many products lines high-value costs such as tables, copier machines, phones, binders, chair, and storage.

Anova: Single Factor						
SUMMARY						
Groups	Count	Sum	Average	Variance		
Consumer	3635	551.0086555	0.151584224	0.015958		
Corporate	3346	503.1997439	0.150388447	0.020161		
Home Office	2922	411.8321292	0.140941865	0.022792		
ANOVA						
Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	0.21133567	2	0.105667835	5.448284	0.0043166	2.996638963
Within Groups	192.007523	9900	0.019394699			
Total	192.218858	9902				

Figure 9: Diagnostic analysis for discount differences among customer segments

Moreover, as presented in Figure 9 above, diagnostic analysis (ANOVA) suggests a significant difference in discounts available for the product among various customer segments, significantly affecting the profit of the superstore. Predictive analysis by regression model and diagnostic analysis done by Danziger et al., (2014) demonstrate that consumers are more likely to buy products from retailers who offer discounted prices every day or continuously. Additionally, customers did not prefer the retailer offering infrequent but high discounts and preferred a retailer offering frequent low discounts. Moreover, price heterogeneity is also affecting the sales revenue of the product. On the other hand, Koçaş & Bohlmann, (2008) analyse that some retailers are giving deep discounts and forgo price-sensitive switchers to earn more profitability within a shorter time and are also incorporating price optimisation benefits.

Managers or retailers should make data-driven decisions to provide lower discounts on different products under different brands rather than promoting high deals irregularly. Moreover, the manager needs to focus on the price optimisation strategy so that one loss-making product can be recovered from the other products and can make trade-offs between price reduction and revenue. Besides this, competitors' pricing and discount strategies significantly affect the existing customer base; thus, the manager can make data-driven decisions for price optimisation and everyday low pricing.

Logistic/Shipping preference for profitability and customer satisfaction

In today's era of digital technology, logistic and transportation services is crucial for retailers and the e-commerce industry. In e-commerce and retail, the most critical aspect is the delivery mode, delivery time, and services in online shipment to the customer. Retail retailers provide door-to-door services to increase customer satisfaction in traditional retail business management. Superstore retailer services differ from e-commerce in terms of delivery promptness, security, product availability, delivery acknowledgment, and product tracking (Haron et al., 2017). For higher profitability and customer retention, the manager must identify the suitable shipping mode and set order priority for better customer service and segmentation. Below is the descriptive analysis for various variables related to online delivery and preferred shipment mode.

Particulars	Sales value	Quantity in order	Discount	Profit	Shipping Cost	Delivery Days
Mean	245.974303	3.476545136	0.142907545	28.36100604	25.9858062	3.969370248
Standard Error	2.152845533	0.01006198	0.00093733	0.769155876	0.252798395	0.007636393
Median	85	3	0	9	7	4
Mode	13	2	0	0	1	4
Standard Deviation	487.5612927	2.278766314	0.212279932	174.1930053	57.25199987	1.729436584
Sample Variance	237716.0142	5.192775913	0.045062769	30343.2031	3277.791489	2.990950899
Kurtosis	176.728322	2.27588873	0.716682408	292.2445931	50.13984133	-0.250010605
Skewness	8.137969532	1.360367731	1.387774552	4.169213311	5.872543775	-0.432053071
Range	22638	13	0.85	14998	932	7
Minimum	0	1	0	-6599	1	0
Maximum	22638	14	0.85	8399	933	7
Sum	12616022	178312	7329.728	1454636	1332812	203589
Count	51290	51290	51290	51290	51290	51290
Largest(1)	22638	14	0.85	8399	933	7
Smallest(1)	0	1	0	-6599	1	0
Confidence Level(95.0%)	4.219599288	0.019721583	0.001837177	1.507553393	0.495487442	0.014967409

Figure 10: Descriptive data analysis

The above table demonstrates the descriptive analysis of the various data variables. It illustrates that 51,290-customer data are in the data set with an average sales value of \$245.97. Furthermore, the average quantity in the order is three units with an average 14% discount rate. Based on this, the manager can analyse products according to shipping mode and order priority to streamline the order delivery system and help manage inventory. Further, the maximum delivery days are 7, and the highest shipping cost is \$933 per order.

Sum of Shipping Cost by Ship Mode and Segment

Segment ● Consumer ● Corporate ● Home Office

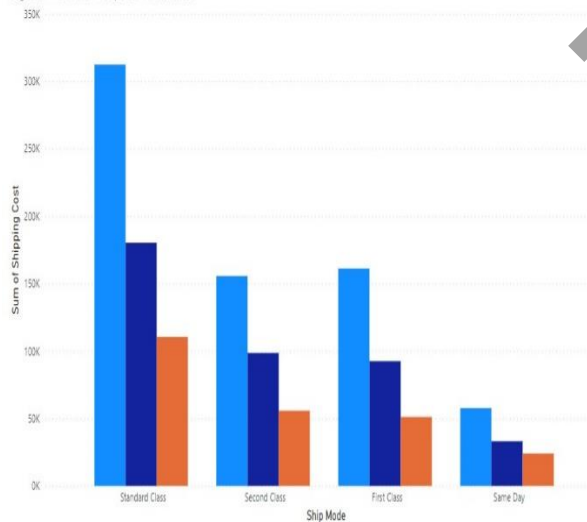


Figure 11: Shipping cost

Count of Delivery Days by Order Priority and Segment

Segment ● Consumer ● Corporate ● Home Office

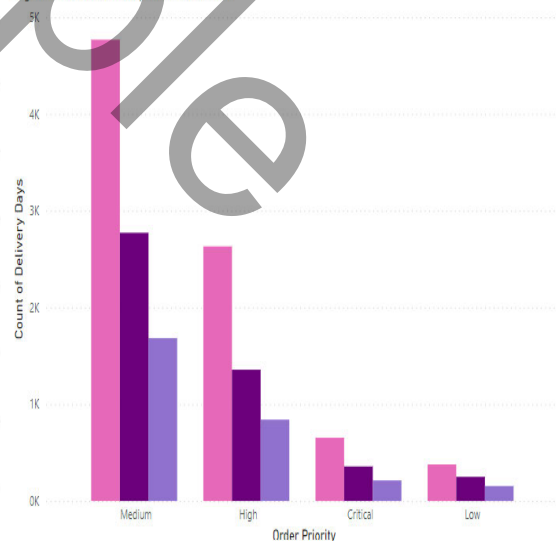


Figure 12: Delivery days

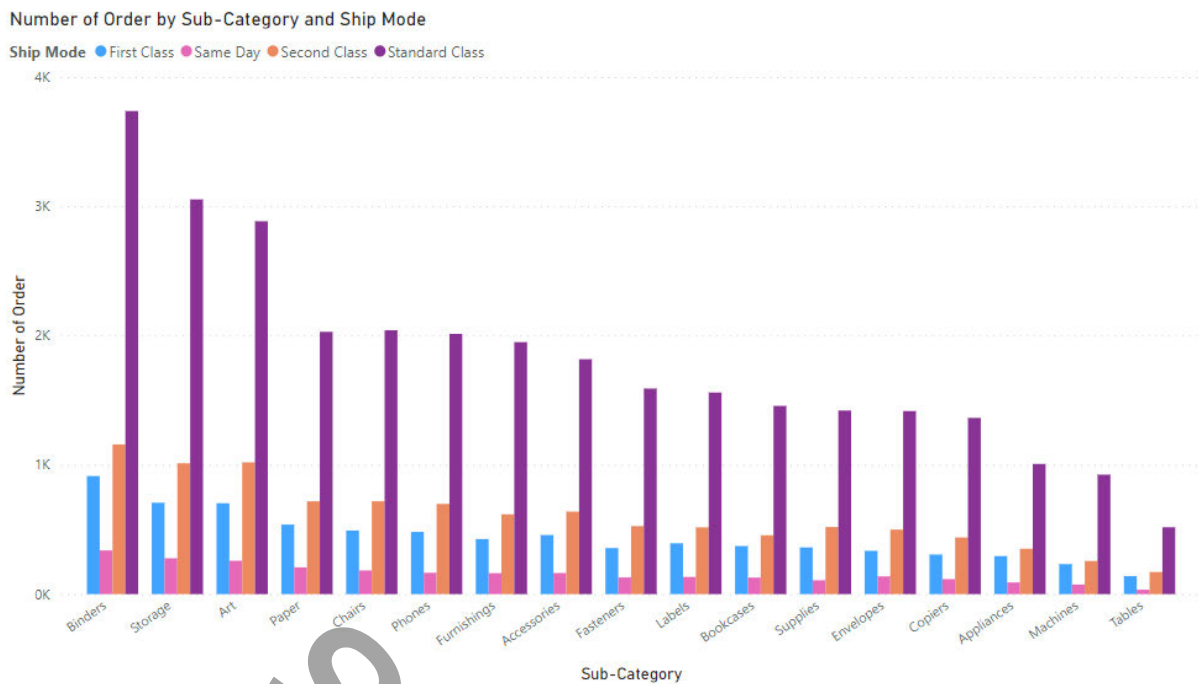


Figure 13: Number of products shipped under different ship modes.

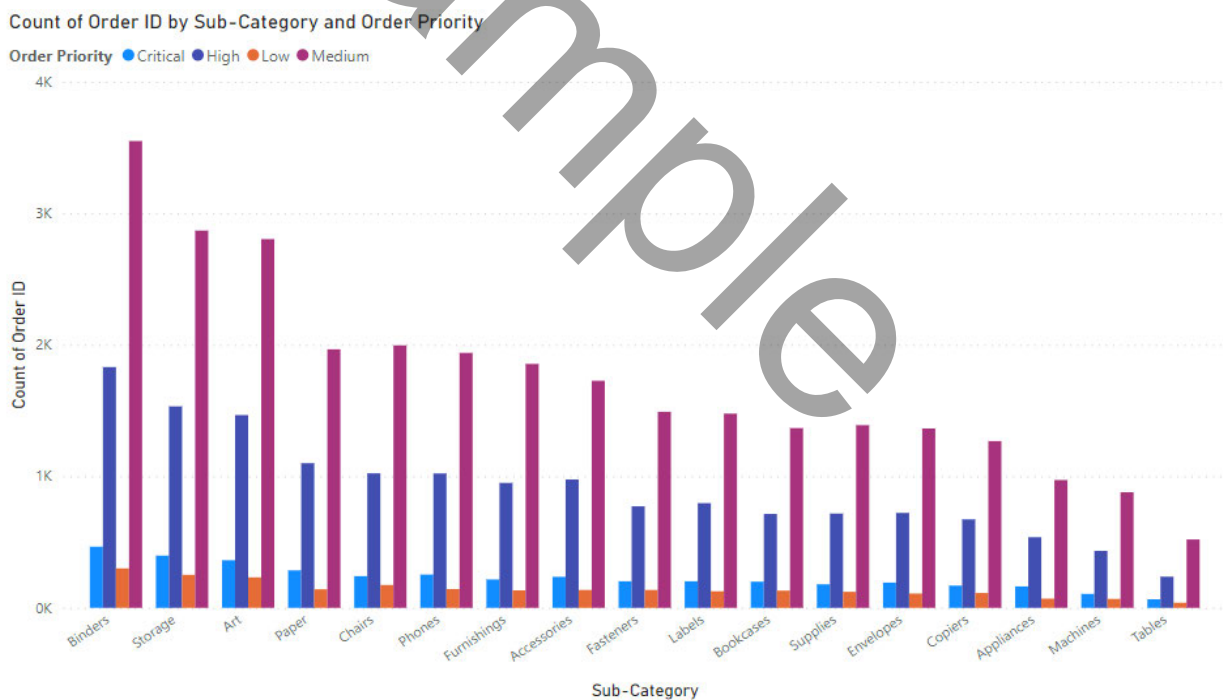


Figure 14: Number of orders shipped as per order priority.

The descriptive analysis in Figure 13 illustrates that the highest number of products from all categories are shipped under standard class, and the lowest is shipped under first class shipment mode. Additionally, as per Figure 11, the standard class shipment mode has the highest shipping cost, followed by second, first, and same-day shipment modes. Besides this, the bar graph in

Figure 14 shows that an increased number of products are delivered as medium order priority, with a relatively high number of delivery days presented in Figure 12. The manager should analyse the whole product line and standardise the shipment mode according to the product because lower sales value order can be shipped under same day or first class shipment to save the cost and helps to decrease delivery days. However, the manager could prioritize the shipment mode per the order value. Retailers also can analyze customer segmentation based on the customer's buying pattern. Predictive analysis was done for the customer segmentation based on the selected data, presented below in Figure 15.

x-Means - Summary

Number of Clusters: 4

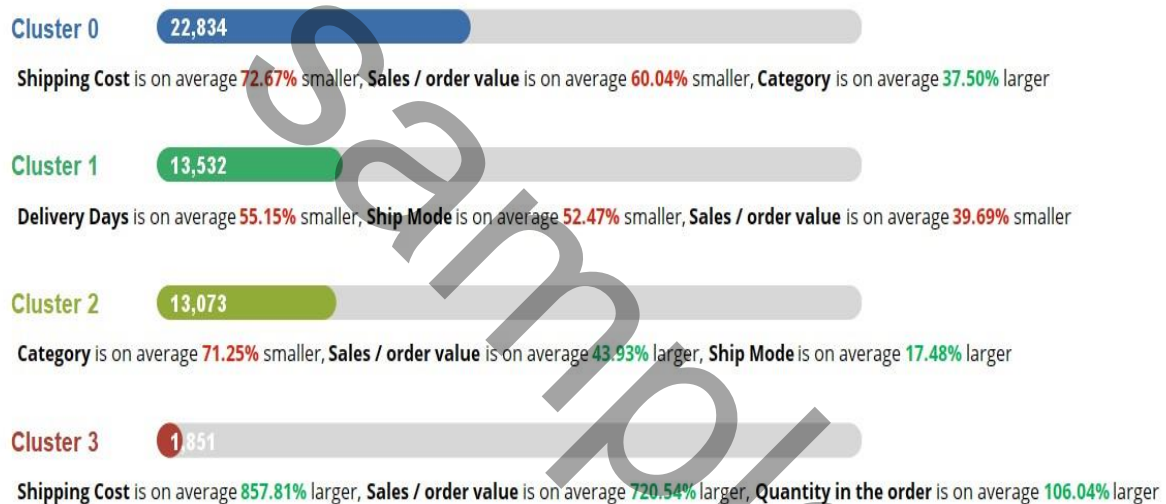


Figure 15: Customer clustering as per product category (RapidMiner)

x-Means - Centroid Table

Cluster	Category	Delivery Days	Discount	Market	Order Pri...	Profit	Quantity...	Region	Sales / or...	Segment	Ship Mode	Shipping ...	Sub-Cate...
Cluster 0	1.942	4.916	0.145	1.929	0.832	9.730	3.415	3	98.289	0.674	2.863	7.828	3
Cluster 1	1.582	1.780	0.149	1.809	1.074	7.128	3.324	19	148.358	0.656	1.116	23.259	3
Cluster 2	0.406	4.647	0.144	1.621	0.817	23.458	3.371	19	354.030	0.659	2.759	30.177	13
Cluster 3	0.747	3.504	0.059	1.611	1.005	448.041	6.103	19	2018.324	0.667	2.103	240.318	13

Figure 16: Centroid table

Customer segmentation is breaking a market into distinct groups of consumers with varying demands, traits, and behaviours who may need different goods or react differently to different marketing strategies. Thus, managers should focus on clustering to gain more profitability from the targeted group of customers. As illustrated in Cluster 0, a total of 22,834 customers bought

office supplies category with lower sales order values which shipping cost is also, on average smaller (Cluster 0 centroid table value is 1.942, which is close to 2 and 2= office supplies as per Figure 1). Moreover, cluster 1 is also an important segment to analyse as first-class shipment mode has, on average, fewer delivery days for products with the lower order value. However, as per cluster 1, the least number of orders are shipped under first-class ship mode with the lower order value. Thus, managers should make data-driven decisions to manage proper inventory for office supply products to fix order priority and shipment mode to increase profitability and customer satisfaction through faster delivery. Additionally, the manager should ship lower-unit-cost products under first-class shipment mode for speedier delivery, which helps increase customer satisfaction and minimize shipping costs.

According to Gümüş et al., (2013), pricing partition for different customers is a common strategy retailers use. From the descriptive and predictive analysis of different variables such as price, shipping cost, retail brands, and customer demographics, Gümüş et al., (2013) analysed two customer segmentation basis in retail. Retailers are likelier to add shipping and handling costs in the lower order value. Besides this, for heavy and fragile products with very high shipping costs, retailers charge shipping costs from the customer. Arguably Cachon et al., (2018) said that e-commerce retailers use a free shipping threshold policy where customers order for a predetermined amount of orders. Regression model validation used and predicted that customers are more likely to order impulse product purchases to increase the threshold limit for free shipping; however, customers are paying less attention to delivery days when they have lower sales order amounts (Cachon et al., 2018). Arguably, Youn et al., (2014) said that profitability and delivery services in retail and e-commerce are steadily declining due to price competition. He also argues that customers are more likely to be concerned about delivery accuracy and time in delivery services rather than higher or lower product delivery costs.

Recommendation

Data-driven decision-making plays a crucial role in this digital era, as much data are available in different industries. For the retail and e-commerce sector, comprehensive data are collected with the help of other technological devices. The manager uses these data to make data-driven decisions for competitive advantage, inventory management, pricing strategy, and shipping preferences for the product and customer segments. It can be recommended for the manager and retailer that low-performing product analysis is significant to identify the cause of the loss for superstore table products. Managers and retailers must identify the reason for losing a

specific product. They should focus on strategies to mitigate the loss by analysing market trends and customer buying patterns. Furthermore, Pricing and discounting strategies are vital in increasing profitability. It is recommended for the retailer that the superstore should promote discounting on different products regularly instead of providing deep discounts on infrequent. Moreover, setting logistic preference shipment mode as per order value and development type increases customer satisfaction by faster delivery days.

Conclusion

This report is designed to analyse the dataset and make data-driven decisions in contemporary business issues in office equipment and product supply retail superstores. Throughout the report, three retail business context data analyses were discussed, such as product analysis, discounting and pricing strategy, and shipping or logistic preference for the superstore with the literature review. According to Kumar et al., (2017), market, firm, store, and customer-level marketing strategies are essential to integrate into the store's management because only high discounts and shipping costs are not causes of the loss in the product line. Besides this, frequent discount on products and price optimisation helps retailers design appropriate pricing strategies for loyal customers (Danziger et al., 2014; Koçuş & Bohlmann, 2008). On the other hand, logistic service and order priority are essential factors for customer segmentation based on order value. Youn et al., (2014) demonstrate that customers are more likely to be concerned about faster delivery time than delivery cost in retail and e-commerce. Overall, managers in various industries can make data-driven decisions for better business management based on different data analysis methods such as descriptive, diagnostic, prescriptive, and predictive techniques.

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